| **UMA028**: **Mathematics for Data Science** | | | | |
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| | 3 | 0 | 2 | 4.0 |

**Course Objective**: To introduce the student to the concept of Probability and Statistics that plays a vital role in computing and computational intelligence. Knowledge of these topics is critical to decision making and to the analysis of data. Using concepts of probability and statistics, individuals are able to predict the likelihood of an event occurring, organize and evaluate data.

**Mathematical Foundations of Data Sciences:** Matrices, Vectors, Vector Spaces, Matrix Decomposition, Singular Value Decomposition, Eigenvalues and vectors, Sets and classes, Limit of a sequence of sets, rings, sigma-rings, sigma fields, monotone classes.

**Probability:** Classical, relative frequency and axiomatic definitions of probability, addition rule and conditional probability, multiplication rule, total probability, Bayes‟ Theorem and independence, Random variable, some common discrete and continuous distributions (Binomial, Poisson, Negative binomial, Geometric, Rectangular, Exponential, Normal, Gamma).

**Bi-variate Probability Distribution:** Probability distribution of functions of a random variable, Joint and marginal distributions, Conditional distributions.

**Correlation and Regression:** Covariance, Karl-Pearson and rank Correlation coefficients; linear regression between two variables.

**Estimation:** Theory of Estimation, Properties of an estimator: Unbiasedness, consistency, Method of maximum likelihood, the method of moments, confidence intervals for parameters in one sample and two sample problems of normal populations, confidence intervals for proportions.

**Hypothesis:** Introduction to Sampling Distribution (standard normal, chi-square, T& F distributions), Critical regions, Neyman-Pearson lemma (without proofs).

**Parametric & Non-parametric tests:** Tests for Goodness of fit: Based on Chi-square Test, one sample and paired sample tests; Sign Test, Signed-rank Test, Kolmogorov Smirnov Test.

**Data Processing:** Regression, Dimensionality Reduction, Linear Discriminant Analysis Principal Component Analysis.

**Laboratory Work:** Lab work based on the programming in MATLAB/ Python /SPSS/R language of various statistical techniques.

**Course Learning Outcomes (CLO)**
After completion of this course, the students will be able to:
1. compute probabilities of composite events along with an understanding of random
2. variables and distribution functions.
3. understand the convergence of sequence in probabilities
4. analyse the correlated data and fit the linear regression models

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| 5. make statistical inferences using principles of hypothesis tests. |
| **Text Books**<br>1. Meyer P. L., Introduction to Probability and Statistical Applications, Oxford & IBH, (2007).<br>2. Hogg, R. V. and Craig, A.T., Introduction to Mathematical Statistics, Prentice Hall of India, (2004).<br>3. Ross, S.M., A First Course in Probability, 9th edition, Pearson (2012).<br>4. Peng, D., R., R Programming for Data Science, Lulu.com (2012). |
| **Reference Books**<br>1. Walpole, R. E., Myers, R. H., Myers, S. L. and Ye, K.,. Probability and statistics for engineers and scientists, Pearson, (2010).<br>2. Hestie Trevor, Tibshirani R., Friedman J., the elements of statistical learning, Springer Verlag New York Inc., 2nd Ed., (2001). |

| **Evaluation Scheme** | |
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| **Evaluation Elements** | **Weightage %** |
| **Mid Semester Test (MST)** | **25-30** |
| **End Semester Examination (ESE)** | **40-45** |
| **Sessional** (may include Assignment, Sessional (Includes Regular Lab assessment )<br>and Quizzes Project (Including report, presentation etc.) | **30** |